

Day 5 - Largest of Three Numbers

Question

Teen numbers diye gaye hain.

Unme se sabse bada (largest) number find karo.

Example

Input

a = 10
b = 25
c = 15

Output

25

Approach 1 - Using Nested if-else

Logic

- Agar a bada hai b aur c dono se $\rightarrow$ a largest
- Agar b bada hai a aur c dono se $\rightarrow$ b largest
- Warna c largest

Dry Run

```
a = 10  
b = 25  
c = 15
```

```
25 > 10 ✓  
25 > 15 ✓
```

```
Largest = 25
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Java Code

```
public class Day5 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 25;  
        int c = 15;  
  
        if(a > b) {  
            if(a > c)  
                System.out.println(a);  
            else  
                System.out.println(c);  
        } else {  
            if(b > c)  
                System.out.println(b);  
            else  
                System.out.println(c);  
        }  
    }  
}
```

Approach 2 - Using Logical AND (&&)

Logic

Directly check karo:

`a > b && a > c`

Java Code

```
public class Day5 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 25;  
        int c = 15;  
  
        if(a >= b && a >= c)  
            System.out.println(a);  
  
        else if(b >= a && b >= c)  
            System.out.println(b);  
  
        else  
            System.out.println(c);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Approach 3 - Using Math.max()

Logic

Pehle a aur b me largest nikalo.

Fir uska c ke saath compare karo.

Java Code

```
public class Day5 {  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 25;  
        int c = 15;  
  
        int largest = Math.max(a, Math.max(b, c));  
  
        System.out.println(largest);  
    }  
}
```

Time Complexity

$O(1)$

Space Complexity

$O(1)$

Approach 4 - Using Array Traversal

Logic

Agar numbers bahut saare ho jaye to array use karte hain.

Java Code

```
public class Day5 {  
    public static void main(String[] args) {  
  
        int[] arr = {10, 25, 15};  
  
        int largest = arr[0];  
  
        for(int i = 1; i < arr.length; i++) {  
            if(arr[i] > largest)  
                largest = arr[i];  
        }  
  
        System.out.println(largest);  
    }  
}
```

Time Complexity

$O(N)$

Space Complexity

$O(1)$

Interview Follow-Up Questions

Q1. Agar teeno equal ho?

```
a = 10  
b = 10  
c = 10
```

Output:

10

Q2. Negative numbers ke liye chalega?

Haan.

```
a = -5  
b = -2  
c = -10
```

Output:

-2

Q3. Interview me best answer?

Pehle:

```
if(a >= b && a >= c)
```

Fir batao:

```
Math.max(a, Math.max(b, c))
```

Java specific shortcut hai.

Interview Concept of the Day

Logical AND (&&)

```
if(a > b && a > c)
```

Matlab:

```
Condition 1 = True  
AND  
Condition 2 = True
```

Dono true honi chahiye.

Example

```
10 > 5 && 10 > 8  
True && True
```

```
Result = True
```

Summary

Approach	Time	Space
Nested if	O(1)	O(1)
Logical AND	O(1)	O(1)
Math.max()	O(1)	O(1)
Array Traversal	O(N)	O(1)

Practice Questions

Predict Output:

1

```
a = 100;  
b = 50;  
c = 25;
```

2

```
a = -10;  
b = -5;  
c = -1;
```

3

```
a = 7;  
b = 7;  
c = 7;
```

4

```
a = 0;  
b = 100;  
c = 50;
```

5

```
a = 999;  
b = 1000;  
c = 998;
```